INCH-POUND

MIL-DTL-83725/23A
29 September 2022
SUPERSEDING
MIL-R-83725/23(USAF)
23 November 1987

DETAIL SPECIFICATION SHEET

RELAYS, VACUUM, SPDT, 15 AMPERES DC OR 60 HZ RMS, 5 KILOVOLTS PEAK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-DTL-83725.

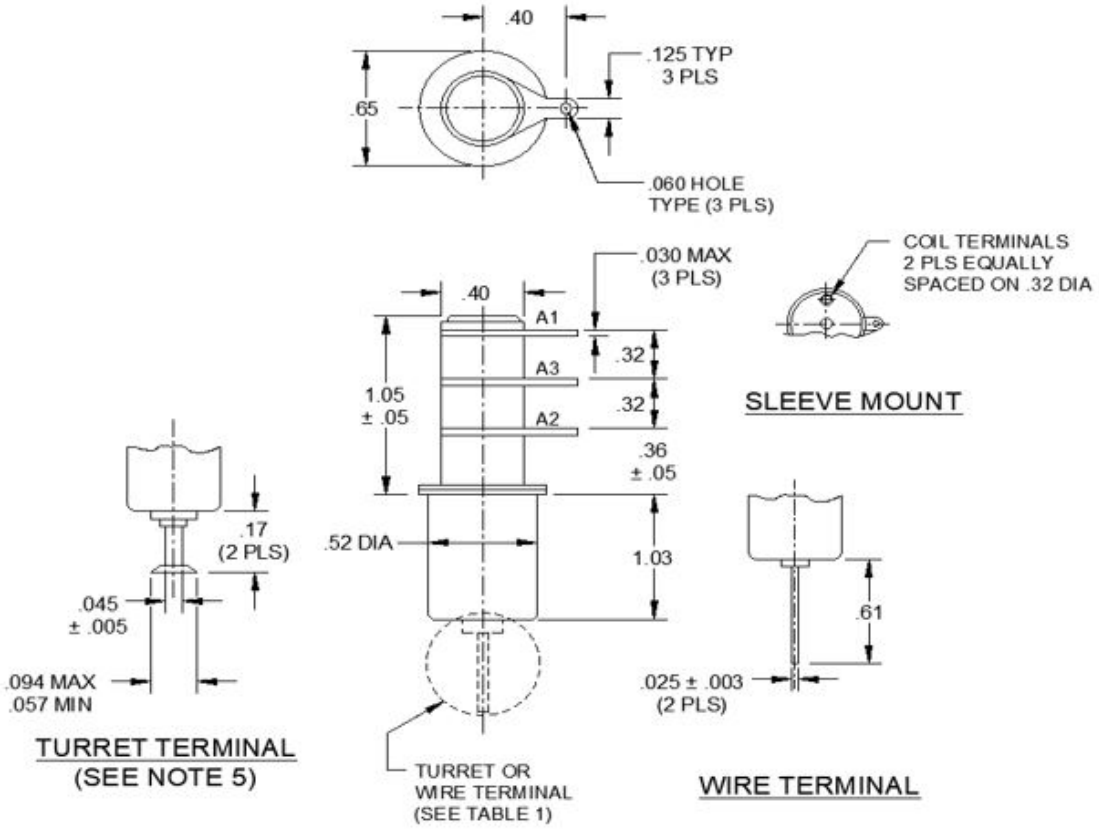
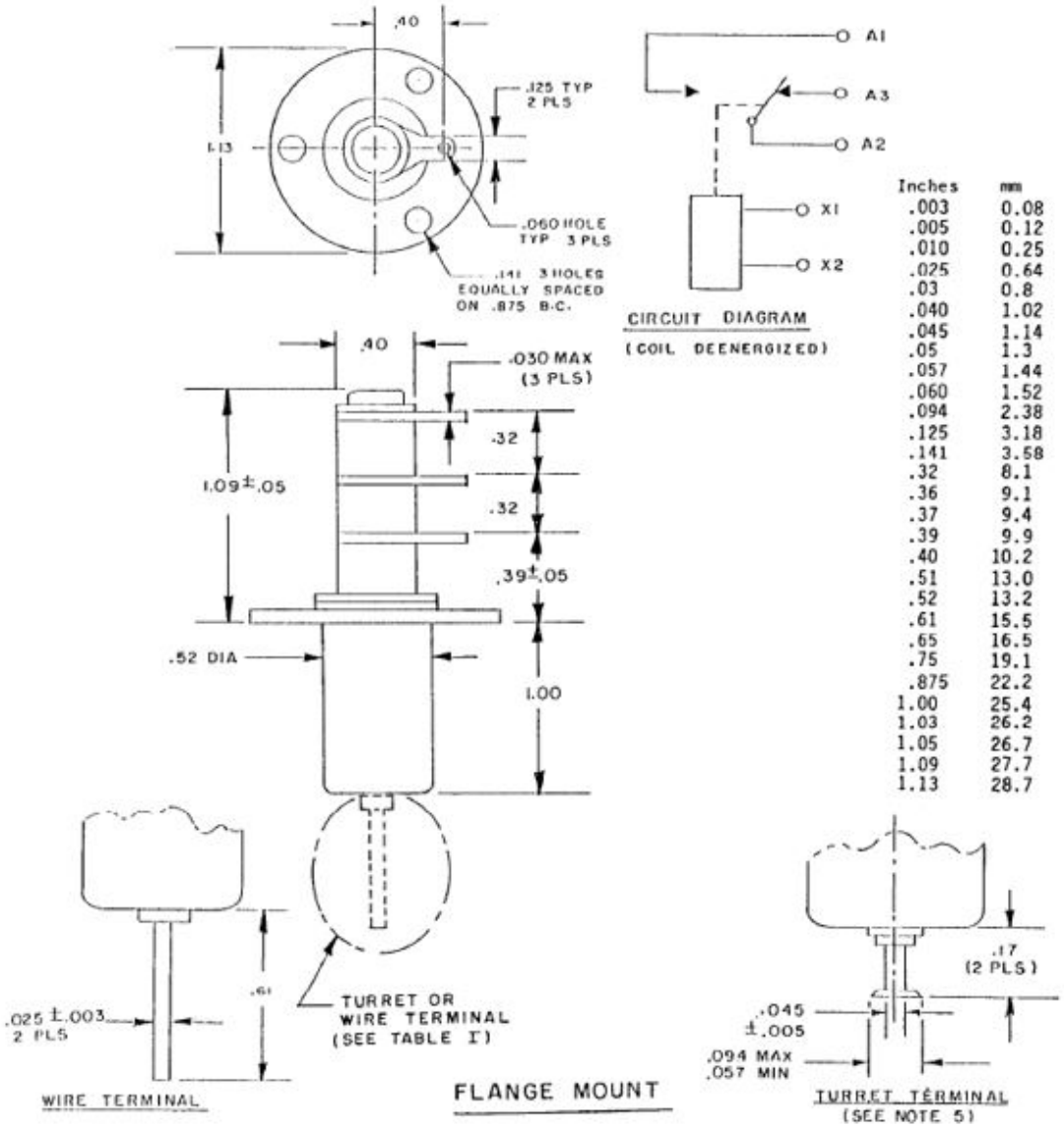


FIGURE 1. Relay dimensions and configuration.

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Inches	mm
.003	0.08
.005	0.12
.010	0.25
.025	0.64
.03	0.8
.040	1.02
.045	1.14
.05	1.3
.057	1.44
.060	1.52
.094	2.38
.125	3.18
.141	3.58
.32	8.1
.36	9.1
.37	9.4
.39	9.9
.40	10.2
.51	13.0
.52	13.2
.61	15.5
.65	16.5
.75	19.1
.875	22.2
1.00	25.4
1.03	26.2
1.05	26.7
1.09	27.7
1.13	28.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.76 mm) for two places decimals and ±.010 (0.25 mm) for three place decimals.
4. Glazing of ceramic insulator is required.
5. Turret terminals shall accommodate two number 22 AWG wires as a minimum.

FIGURE 1. Relay dimensions and configuration - Continued.

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REQUIREMENTS:

CONTACT DATA:

Configuration: SPDT, ground isolated.
Arrangement: 1 form C.
Load ratings:
Resistive: (Carry only).
15 amperes dc or 60 Hz rms.
12 amperes rms, 2.5 Hz.
8 amperes rms, 16 MHz.
6 amperes rms, 32 MHz.
2.75 amperes rms, 50 MHz.
2.0 amperes rms, 75 MHz.

Rated operating voltage:
5 kilovolts peak, 60 Hz or dc.
4.5 kilovolts peak, 2.5 MHz.
3.5 kilovolts peak, 16 MHz.
2.8 kilovolts peak, 32 MHz.
2.0 kilovolts peak, 50 Hz.
1.0 kilovolt peak, 75 MHz.

Contact resistance:
Rated Life:
Before: .020 ohm, maximum.
During: .100 ohm, maximum.
After: .100 ohm, maximum.

Capacitance:
1.6 picofarads, maximum across open contacts.
1.6 picofarads, maximum between open contacts and ground.

COIL DATA: (See table I).

Duty rating: Continuous.
Operate time: 10 milliseconds, maximum, including bounce.
Release time: 10 milliseconds, maximum, including bounce.

ELECTRICAL DATA:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and case at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure, 60 Hz or dc.
Between all mated contacts in the open position: 6 kV peak.
Between high voltage terminals and case: 6 kV peak.
Between coil and case: 500 V rms, 60 Hz.

ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.
Vibration: MIL-STD-202-204, test condition C (10-55 Hz, .06-inch DA/55-2000 Hz, 10 g).
Shock: MIL-STD-202-213, test condition A, 50 g, 11 milliseconds, half sine.

PHYSICAL:

Terminal strength: 5 pounds pull.
Dimensions and configuration: See figure 1.
Termination: Solder terminal.
Weight: 1 ounce, maximum.

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LIFE TEST REQUIREMENTS:

Mechanical cycling: 2,000,000 cycles. 2 sample units (cycling rate, 36,000 per hour maximum; contact current shall not exceed 10 milliamperes).

QUALITY ASSURANCE:

Dielectric withstanding voltage: Tests to be conducted at atmospheric pressure rating only.
Duration of application: 5-10 seconds at a 10 percent increase in the dielectric withstanding voltage.

PART NUMBER: M83725/23- and a dash number from table I.

TABLE I. Dash numbers and characteristics. 1/

Part number M83725/23-	Mount	Coil terminal type	Coil voltage rated (V dc) 2/	At 25° C 3/					Over temperature range		
				Coil resistance (ohms ±10%)	Coil voltage maximum (V dc)	Pickup voltage maximum (V dc)	Hold voltage (V dc)	Drop-out voltage (V dc)	Pickup voltage maximum	Hold voltage (V dc)	Dropout voltage (V dc)
001	Sleeve	Wire	12	70	14	8	5	0.5	11	7	0.3
002		"	26.5	290	32	16	10	1	23	14	0.7
003	"	"	115	4700	127	80	50	5	110	70	3.5
004	Flange	"	12	70	14	8	5	0.5	11	7	0.3
005	"	"	26.5	290	32	16	10	1	23	14	0.7
006	"	"	115	4700	127	80	50	5	110	70	3.5
007	Sleeve	Turret	12	70	14	8	5	0.5	11	7	0.3
008	"	"	26.5	290	32	16	10	1	23	14	0.7
009	"	"	115	4700	127	80	50	5	110	70	3.5
010	Flange	"	12	70	14	8	5	0.5	11	7	0.3
011	"	"	26.5	290	32	16	10	1	23	14	0.7
012	"	"	115	4700	127	80	50	5	110	70	3.5

1/ Relay case must be grounded.
2/ CAUTION: The use of any coil voltage other than rated coil voltage may compromise the operation of the relay.
3/ Energizing the coil for 30 minutes prior to measurement of the pick-up voltage is not applicable.

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QUALIFICATION INSPECTION:

Qualification inspection: See table II.

TABLE II. Qualification inspection.

Single submission	Group submission		
	Part	Sample size	Inspections
10 units plus 1 open unit	M83725/23-005	5 units	Full qualification
	M83725/23-011	5 units	
	M83725/23-001	1 unit	Group I of qualification inspection table, shock, vibration, plus 1 open unit of any dash number
	M83725/23-004	1 unit	
	M83725/23-009	1 unit	
	M83725/23-012	1 unit	

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-83725, this document references the following:
MIL-STD-202-204 MIL-STD-202-213

Custodians:
Air Force - 85
DLA - CC

Preparing activity:
DLA-CC

(Project 5945-2022-046)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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